

Enthusiast Course

PHASE : Speed & Accuracy Test

TARGET : PRE-MEDICAL 2022

Test Type : PRACTICE TEST

Test Pattern : NEET (UG)

TEST DATE :

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	3	3	3	2	3	1	3	3	2	4	3	2	4	1	4	3	3	1	4	2	2	2	4	2	1	3	2	3	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	3	3	2	3	3	4	2	2	1	2	2	2	1	2	3															



HINT - SHEET

1. Ans (1)

$$m = \frac{m_0}{e^{t/T_{av}}} \quad \text{Given } t = 2T_{av}$$

$$m = \frac{40}{e^2} = \left(\frac{1}{e}\right)^2 \times 40 = 0.137 \times 40$$

$$m \simeq 5.40 \text{ g}$$

2. Ans (3)

$$1 \text{ week} \simeq 7 \text{ days} \simeq 7 \times 24 \text{ hour} \simeq 14 \text{ half life}$$

$$\text{Number of atoms left } \frac{N_0}{2^{14}}, \text{ Activity} = N\lambda$$

$$\therefore \text{Activity left is } \frac{1}{2^{14}} \text{ times the initial} = \frac{1}{2^{14}} \times 1 \text{ curie}$$

$$= \frac{1}{16384} \times 1 \text{ curie} \simeq 61 \times 10^{-6} \text{ curie} \simeq 60 \mu \text{ curie}$$

3. Ans (3)

$$N_0 = 200 \text{ gm} \quad N = 50 \text{ gm} = \frac{N_0}{4} = \frac{N_0}{2^2}$$

$$\therefore \frac{t}{T} = 2 \Rightarrow t = 2T = 3200 \text{ yr}$$

4. Ans (3)

From Einstein's photo electric effect equation.

$$\frac{hc}{\lambda} = \phi_0 + \frac{1}{2}mv^2 \quad \dots(1)$$

$$\frac{8hc}{7\lambda} = \phi_0 + \frac{1}{2}mv_1^2 \quad \dots(2)$$

$$\Rightarrow \frac{8}{7}(\phi_0 + \frac{1}{2}mv^2) = \phi_0 + \frac{1}{2}mv_1^2$$

$$\Rightarrow \frac{1}{2}mv_1^2 = \frac{\phi_0}{7} + \frac{1}{2}m\left(\sqrt{\frac{8}{7}}v\right)^2$$

$$\Rightarrow v_1 > \sqrt{\frac{8}{7}}v$$

5. Ans (2)

From Einstein's equation of P.R.E.

$$h\nu = \frac{1}{2}mv^2 + \phi \quad \dots(1)$$

$$2h\nu = \frac{1}{2}mv_1^2 + \phi \quad \dots(2)$$

$$2\left(\frac{1}{2}mv^2 + \phi\right) = \frac{1}{2}mv_1^2 + \phi$$

$$\frac{1}{2}mv_1^2 = \frac{1}{2}m(\sqrt{2}v)^2 + \phi$$

$$v_1 > \sqrt{2}v$$

6. Ans (3)

$$E_k = \frac{hc}{\lambda} - \phi = \frac{12400 \text{ eV} \cdot \text{\AA}}{2000 \text{\AA}} - 4.2 \text{ eV}$$

$$E_k = 6.2 \text{ eV} - 4.2 \text{ eV} = 2 \text{ eV}$$

7. **Ans (1)**

$$E_k = h\nu - \phi$$

$$= (4.13 \times 10^{-15} \text{ eV.s})(5 \times 10^{14} \text{ s}^{-1}) - 1.5 \text{ eV}$$

$$= 2.065 \text{ eV} - 1.5 \text{ eV}$$

$$\approx 0.5 \text{ eV}$$

8. **Ans (3)**

$$h = 6.63 \times 10^{-34} \text{ J.s} = 6.63 \times 10^{-34} \text{ kg m}^2/\text{s}$$

$$= 4.13 \times 10^{-15} \text{ eV.s}$$

9. **Ans (3)**

increasing frequency order

$$\nu_{\text{(infrared ray)}} < \nu_{\text{ultra violet rays}} < \nu_{\text{x-rays}} < \nu_{\gamma\text{-rays}}$$

10. **Ans (2)**

Energy of photon > Threshold energy

11. **Ans (4)**

$$n_e \propto \frac{1}{d^2}$$

12. **Ans (3)**

$$p = \frac{h}{\lambda} = \text{constant} (\because \lambda = \text{constant})$$

Energy of photon $E_p = p.c.$

$$\text{Kinetic energy of electron } E_k = \frac{1}{2}mv^2 = \frac{1}{2}pv$$

$$\frac{E_p}{E_k} = \frac{pc}{\frac{1}{2}pv} = \frac{2c}{v} > 1 \quad (\because c > v)$$

$$E_p > E_k$$

13. **Ans (2)**

$$\lambda = \frac{h}{\sqrt{2mE}} \quad \text{or} \quad \lambda \propto \frac{1}{\sqrt{m}}$$

14. **Ans (4)**

$$\lambda = \frac{h}{\sqrt{2mE_k}} \quad \text{or} \quad \lambda \propto \frac{1}{\sqrt{m}}$$

15. **Ans (1)**

$$\lambda = \frac{h}{m_1 v_1} = \frac{h}{m_2 v_2}$$

$$\therefore \frac{v_1}{v_2} = \frac{m_2}{m_1} = \frac{4}{1}$$

16. **Ans (4)**

$\therefore$ Number of active nuclei at instant 't'

$$N = \frac{N_0}{2^{t/Th}}$$

$$\therefore \frac{N_1}{N_2} = \frac{N_0 2^{t/Th_2}}{N_0 2^{t/Th_1}} = \frac{2^{60/20}}{2^{60/30}} = \frac{2^3}{2^2} = \frac{2}{1}$$

17. **Ans (3)**

$$\therefore N = \frac{N_0}{2^{t/Th}}$$

$$\therefore \frac{N_1}{N_2} = \frac{N_{01}}{N_{02}} \times \frac{2^{t/Th_2}}{2^{t/Th_1}} = \left(\frac{2}{1}\right) \left(\frac{2^{24/8}}{2^{24/12}}\right)$$

$$= \frac{2}{1} \times \frac{2^3}{2^2} = \frac{4}{1}$$

18. **Ans (3)**

Total energy of electron = K.E. + Rest mass energy

$$\text{K.E.} = \text{T.E.} - \text{RME}$$

$$= 2.854 \text{ MeV} - 0.511 \text{ MeV} = 2.343 \text{ MeV}$$

19. **Ans (1)**

Total energy = K.E. + Rest mass energy

$$= 0.511 \text{ MeV} + 3.045 \text{ MeV} = 3.556 \text{ MeV}$$

20. **Ans (4)**

$$E = m \times c^2$$

$$E = \frac{9.1 \times 10^{-31} \times 9 \times 10^{16}}{1.6 \times 10^{-19}} \text{ eV}$$

$$E = 0.511 \text{ MeV}$$

21. **Ans (2)**

Mass number decrease by 4 when one α -particle emit but mass number does not change with β -particle.

22. **Ans (2)**

$$N_\alpha = \frac{A_i - A_f}{4} = \frac{232 - 208}{4} = 6$$

$$\text{and } N_\beta = Z_f - Z_i + 2N_\alpha$$

$$N_\beta = 82 - 90 + (2 \times 6) = 4$$

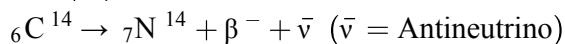
23. **Ans (2)**

$$Th = \frac{0.693}{\lambda} = \frac{0.693}{0.009} = 77 \text{ day}$$

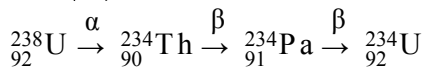
24. **Ans (4)**

$$T_{av} = \frac{Th}{0.693} = 1.44 Th = 1.44 \times 100 = 144 \text{ year}$$

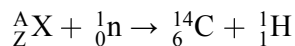
25. **Ans (2)**



26. **Ans (1)**

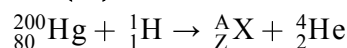


27. **Ans (3)**



Balancing atomic number and mass number on both side we get $Z = 7$, $A = 14$.

28. **Ans (2)**



Balancing atomic number and mass number on both side we get $Z = 79$, $A = 197$.

29. **Ans (3)**

$$m = \frac{m_0}{2^{t/\text{Th}}} \quad \text{or} \quad 2^{t/\text{Th}} = \frac{m_0}{m}$$

$$2^{t/15} = \frac{128}{1} = 2^7$$

$$t = 15 \times 7 = 105 \text{ hour}$$

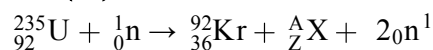
30. **Ans (3)**

$$N = N_0 \left(\frac{1}{2} \right)^{t/\text{Th}}$$

$$\Rightarrow 1 = 16 \left(\frac{1}{2} \right)^{2/\text{Th}}$$

$$\Rightarrow \text{Th} = \frac{1}{2} \text{ hour}$$

31. **Ans (3)**



Atomic number and mass number balance on both side we get $Z = 56$, $A = 142$

32. **Ans (3)**

Atomic number and mass number balance on both side we get $Z=90$, $A=234$.

33. **Ans (2)**

Let the percentage of ${}_{5}^{10}\text{B}$ in sample be x . Then percentage of ${}_{5}^{11}\text{B}$ is $(100-x)$ so using formula of average atomic masses of isotopes

$$10.811 = \frac{(10.01294)x + 11.00931(100 - x)}{100}$$

$$1081.1 = 1100.931 - 0.99637x$$

$$0.99637x = 19.831$$

$$x = \frac{19.831}{0.99637} = 19.9 \simeq 20\%$$

34. **Ans (3)**

Average atomic mass

$$= \frac{75.4 \times 34.98 + 24.6 \times 36.98}{100}$$

$$= 35.47 \text{ u}$$

39. **Ans (1)**

Mass defect (Δm)

$$= [26 m_p + 30 m_n] - (\text{Mass of nucleus } {}_{26}^{56}\text{Fe})$$

$$= [26 \times 1.00783 + 30 \times 1.00867] - [55.93493]$$

$$\simeq 0.52875 \text{ u}$$

$$\text{Binding energy} = \Delta m \times 931 \text{ MeV}$$

$$= 0.52875 \times 931 \text{ MeV}$$

$$\simeq 492.26 \text{ MeV}$$

40. **Ans (2)**

$$\text{BE} = \Delta m \cdot c^2$$

44. **Ans (2)**

$$\frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2} \quad \text{or} \quad mv^2 = \frac{e^2}{4\pi\epsilon_0 r}$$

$$\text{KE} = \frac{1}{2}mv^2 = \frac{e^2}{8\pi\epsilon_0 r}$$

45. **Ans (3)**

$$\frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$$

$$\Rightarrow r = \frac{e^2}{4\pi\epsilon_0 mv^2}$$